

Machine Learning Model Evaluation & Deployment Report

Dynamic Delivery Time Prediction System — Weeks 07 & 08 Deliverable

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1. Executive Summary

We developed and compared four supervised machine learning regression architectures to predict delivery duration (in days) across 96,460 orders. Models were validated using 5-fold cross-validation and hyperparameter grid search.

2. Model Comparison Table

Model	Train MAE	5-Fold CV MAE	Test MAE	Test RMSE	Test R ²
Naive Baseline	6.41 d	6.41 +/- 0.04	6.43 d	9.74 d	0.000
Ridge Regression	5.10 d	5.11 +/- 0.04	5.10 d	8.41 d	0.254
Random Forest (50t)	4.41 d	4.77 +/- 0.03	4.71 d	8.15 d	0.299
XGBoost (Baseline)	4.60 d	4.80 +/- 0.05	4.75 d	8.15 d	0.299
Tuned XGBoost (Final)	4.31 d	4.62 +/- 0.04	4.56 d	8.00 d	0.325

3. Error Analysis & Segment Performance

- Same-State Orders: MAE is 3.05 days (mean actual: 7.83d).
- Cross-State Orders: MAE is 5.42 days (mean actual: 15.22d).
- Extreme Errors: 94.8% of top 1% errors belong to cross-state routes headed to distant northern states (BA, RJ, MA, PA).

4. Production Serialization & Web App

The final pipeline was serialized to `models/final_model.pkl` (987 KB) and integrated into a real-time Streamlit decision support interface (`app/app.py`) providing delivery estimates, promised windows, and operational risk alerts.